

EVEN SUBDIVISIONS ARE CONTRACTIBLE

ABSTRACT. We prove that a bipartite graph is contractible in the sense of graph schemes whenever every vertex in one part has degree at most two. Consequently, replacing every edge of a loopless multigraph by a path of positive even length produces a contractible graph. The proof projects a coloured scheme onto a family of graphic matroids. Each edge label occurs in at most two projections, and a component count verifies the matroid union inequalities needed for simultaneous disjoint spanning trees. If the degrees in that part are at most three, we instead obtain a minor retaining all prescribed roots in the opposite part. Its proof combines the same packing argument with a strictly decreasing root-relocation induction. In particular, every $K_{3,n}$ is weakly contractible.

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1. STATEMENT

All graphs are finite. Target and scheme graphs are simple; auxiliary multigraphs may have parallel edges but no loops. Following Kündgen, Pelsmajer and Ramamurthi [3, Definitions 1.1 and 2.1], an H -scheme in a graph G with $V(H) \subseteq V(G)$ is a family $(P_{uv} : uv \in E(H))$ of paths satisfying two conditions:

- (1) P_{uv} joins u to v and contains no other vertex of H ;
- (2) whenever a nonempty collection of scheme paths has a common vertex, their corresponding edges in H have a common endpoint.

A *rooted H minor* in G is a family $(C_v : v \in V(H))$ of pairwise disjoint connected vertex sets, with $v \in C_v$ for every v , such that C_u and C_v are adjacent whenever $uv \in E(H)$. The graph H is *contractible* if every graph containing an H -scheme contains a rooted H minor.

Theorem 1.1. *Let H be bipartite with a specified bipartition (A, B) . If $d_H(b) \leq 2$ for every $b \in B$, then H is contractible.*

Corollary 1.2. *Let F be a loopless multigraph. Replace each edge of F by a path of positive even length, using mutually disjoint new internal vertices. The resulting simple graph is contractible.*

Proof. Place every original vertex, and every even-position internal vertex of each replacement path, in A . Place the odd-position internal vertices in B . Every vertex of B has degree two, so Theorem 1.1 applies. Isolated vertices of F may be retained. $\square$

Theorem 1.1 includes $K_{2,n}$ for every n , with the part of order n as B . Corollary 1.2 includes the graph obtained by subdividing every edge of any simple graph exactly once. Throughout, “even” refers to the length of each replacement path.

2. PACKING SPANNING TREES WITH SHARED LABELS

Lemma 2.1. *Let J be a finite index set. For each $j \in J$, let M_j be a connected multigraph with a distinguished vertex r_j and an edge set labelled bijectively by a finite set E_j . Suppose that M_j is the union of a family $\mathcal{Q}_j$ of paths whose edge sets partition E_j , all these paths contain r_j , and every other vertex belongs to at least two of them. Suppose also that each element of $E = \bigcup_j E_j$ belongs to at most two sets E_j . Then there are pairwise disjoint sets $T_j \subseteq E_j$ such that T_j is a spanning tree of M_j . Trivial paths are allowed; an empty family is allowed when $V(M_j) = \{r_j\}$.*

Proof. Fix $X \subseteq E$. Let $c_j(X)$ count the components, including isolated vertices, of the spanning subgraph of M_j with edge labels $E_j \cap X$. Put $m_j = |\mathcal{Q}_j|$. The component containing r_j meets all m_j paths, and every other component meets at least two. Conversely, a path Q meets at most $1 + |E(Q) \setminus X|$ components: deleting those edges leaves that many path pieces, each contained in one component. Thus

$$m_j + 2(c_j(X) - 1) \leq \sum_C |\{Q \in \mathcal{Q}_j : V(Q) \cap V(C) \neq \emptyset\}| \leq m_j + |E_j \setminus X|.$$

The last bound uses the partition of E_j into the path edge sets. It follows that

$$(1) \quad c_j(X) - 1 \leq \frac{1}{2}|E_j \setminus X|.$$

Extend the graphic matroid of M_j to the common ground set E by declaring all labels outside E_j to be matroid loops. Its rank function ρ_j satisfies

$$\rho_j(E) = |V(M_j)| - 1, \quad \rho_j(X) = |V(M_j)| - c_j(X).$$

Summing (1) and using the label multiplicity bound gives

$$(2) \quad \sum_j (\rho_j(E) - \rho_j(X)) \leq \frac{1}{2} \sum_j |E_j \setminus X| \leq |E \setminus X|.$$

The finite matroid union rank formula [2] states that

$$\rho_{\bigvee_j \mathcal{M}_j}(E) = \min_{X \subseteq E} \left(|E \setminus X| + \sum_j \rho_j(X) \right).$$

Equation (2), together with $X = E$, makes this rank equal to $\sum_j \rho_j(E)$. Write a maximum independent set of the union as $\bigcup_j I_j$, where each I_j is independent in $\mathcal{M}_j$. Equality throughout

$$|\bigcup_j I_j| \leq \sum_j |I_j| \leq \sum_j \rho_j(E)$$

forces the I_j to be pairwise disjoint bases. A base contains no matroid loops. Therefore $T_j = I_j \subseteq E_j$ is the required spanning tree. $\square$

3. PROOF OF THEOREM 1.1

By the coloured-scheme reduction [3, Lemma 3.3], an H -scheme has a rooted minor whose underlying graph G supports a *coloured H -scheme*. This consists of a proper colouring $f : V(G) \rightarrow V(H)$ fixing each root, such that P_{uv} uses only colours u, v and every nonroot has degree at least four. We remove isolated roots before this reduction and restore them as singleton branch sets at the end. The graph with no remaining roots is immediate.

The paths alternate colours, every edge belongs to exactly one scheme path, and each nonroot lies on at least two scheme paths [3, Remark 3.2(1), (2), (6), (7)]. Set

$$A_a = f^{-1}(a) \quad (a \in A), \quad L_b = f^{-1}(b) \quad (b \in B).$$

If $d_H(b) \leq 1$, then $L_b = \{b\}$, since no nonroot of colour b can lie on two distinct paths incident with b . If $d_H(b) = 2$, every nonroot of colour b lies internally on both paths incident with b . Put

$$E = \bigcup_{b \in B} (L_b \setminus \{b\}), \quad E_a = \bigcup_{b \in N_H(a)} (L_b \setminus \{b\}).$$

These labels are actual vertices of G . Each label belongs to exactly two sets E_a , one for each neighbour of its degree-two root colour.

Construct M_a on vertex set A_a as follows. For each $b \in N_H(a)$ and $x \in L_b \setminus \{b\}$, put an auxiliary edge labelled x between the two neighbours of x on P_{ab} . Both neighbours have colour a , and they are distinct because P_{ab} is a path. Different labels may give parallel edges. Suppress all such vertices x on P_{ab} and remove its terminal vertex b . This leaves a path Q_b^a in M_a from a to the A_a -neighbour of b ; it is trivial if $P_{ab} = ab$. Its edge-label set is exactly $L_b \setminus \{b\}$.

As b ranges over $N_H(a)$, the paths Q_b^a partition E_a , cover A_a , and all contain a . In particular, M_a is connected. Every vertex of $A_a \setminus \{a\}$ lies internally on at least two distinct scheme paths incident with a , so it remains on at least two projected paths. Hence the family $(M_a : a \in A)$ satisfies Lemma 2.1. Choose its pairwise disjoint spanning-tree label sets T_a .

In the reduced graph G , define

$$C_a = A_a \cup T_a \quad (a \in A), \quad C_b = \{b\} \quad (b \in B).$$

These sets are pairwise disjoint: the A_a are different colour classes, the T_a are disjoint sets of nonroot B -colour vertices, and no T_a contains a root. Each tree edge labelled x lifts to the two-edge path through x in G , so $G[C_a]$ is connected. Moreover $a \in C_a$. For each $ab \in E(H)$, the last edge of P_{ab} joins the root b to a vertex of $A_a \subseteq C_a$. Thus these branch sets form a rooted H -minor model.

Finally compose this model with the rooted minor reduction. Replace every vertex in each branch set by its connected preimage in the original graph. Disjointness, connectivity, root containment and required adjacencies are preserved. Restoring the isolated roots completes the proof. The singleton assertion for the B branch sets applies only in the reduced coloured scheme. $\square$

4. DEGREE THREE WITH ONE PART OF THE ROOTS RETAINED

An H -minor model is *rooted on* $A \subseteq V(H)$ if $a \in C_a$ for every $a \in A$; roots outside A need not be retained. A target is *weakly contractible* if every scheme of it contains an unrooted minor of it.

Lemma 4.1. *Let H be bipartite with bipartition (A, B) . If every nonroot of a B colour in a coloured H -scheme belongs to exactly two scheme paths, then the underlying graph contains a fully rooted H minor.*

Proof. Repeat the projections in Section 3, but include a label x in M_a only when x actually occurs on P_{ab} . Each label then belongs to exactly two projections, even when its target colour has degree greater than two. For fixed a , the projected paths partition the labels, cover $f^{-1}(a)$, and all contain a . Every other vertex belongs to at least two projected paths. Lemma 2.1 therefore supplies disjoint spanning-tree label sets. The same branch sets $C_a = f^{-1}(a) \cup T_a$, $C_b = \{b\}$ give the rooted model: the last edge of each P_{ab} still witnesses its required contact. Isolated roots are singletons. $\square$

Theorem 4.2. *Let H be bipartite with bipartition (A, B) and $d_H(b) \leq 3$ for every $b \in B$. Every H -scheme contains an H minor rooted on A . Consequently H is weakly contractible.*

Proof. Delete isolated target roots from the host and restore them as their original singleton branch sets at the end. They lie on no scheme path; an edgeless target is immediate. For the remaining fixed target, use strong induction on host order. During this argument regard H as an abstract target with an injective current root map $v \mapsto r_v$.

The reduction in [3, Lemma 3.3] gives a root-preserving minor K supporting a coloured scheme, with $|V(K)| \leq |V(G)|$. Suppose a nonroot x of colour $b \in B$ lies on every path incident with b . Move the root of b to x and replace each P_{ba} by its suffix from x to r_a . No other path contains x , since its colour is b . The new paths avoid the old root r_b , contain no other current root internally, and inherit the scheme intersection condition from the old paths. They form an H -scheme in $K - r_b$, whose order is strictly smaller than $|V(G)|$. The induction hypothesis gives a model rooted on A , since none of those roots moved. Its lift to G retains every original A root.

Otherwise, every nonroot of a B colour has exactly two memberships. Indeed each has at least two; a degree-one colour admits none, a degree-two colour would have full support, and three memberships at a degree-three colour would also give the preceding case. Apply Lemma 4.1 and lift the

resulting model to G . Every recursive step strictly lowers host order, and every lift preserves the required A roots. $\square$

5. SCOPE

The contractible targets supplied by Corollary 1.2 have unbounded treewidth, since they include a once-subdivision of K_t for every t . This does not assert that contractibility is minor-closed or that K_t itself is contractible. Theorem 1.1 also covers theta graphs whose three paths all have even length; it does not cover the case of three odd-length paths. It therefore addresses only part of the bipartite-theta question in [3, Section 8, Question 3]. Theorem 4.2 gives a $K_{3,n}$ minor retaining its three prescribed roots in the part of order three. For $K_{3,3}$ it does not guarantee retention of all six roots; its unrooted special case is already [3, Theorem 5.3]. Neither theorem establishes full bipartite contractibility or a Hadwiger conjecture.

Biswal, Lee and Rao [1, Lemma 3.2] state a broader unrooted claim for integral unit flows with bipartite demand of minimum degree two. We use their intended preprint-v2 intersection convention: paths with four distinct demand endpoints are vertex-disjoint when the independent intersection count is zero. The published definition appears to reverse “intersect” and “are vertex disjoint”; the following issue concerns the intended convention, independently of that apparent typo.

Their supplied prefix construction fails even for the coloured $K_{2,2}$ -scheme on eight distinct vertices with paths

$$P_{ij} = a_i b'_j a'_i b_j \quad (i, j \in \{0, 1\}).$$

Each left prefix stops strictly before its first vertex on a path incident with another left root, giving $C_{a_i} = \{a_i\}$ and $C_{b_j} = \{b_j, b'_j, a'_0, a'_1\}$. The right sets overlap. This refutes the construction, including the containment assertion of their Lemma 3.6, not the intended existence statement of Lemma 3.2. The graph does have the rooted model $D_{a_i} = \{a_i\}$, $D_{b_j} = \{b_j, b'_j, a'_j\}$. Our proofs do not use that broader claim; publication priority remains to be assessed in light of it.

REFERENCES

- [1] P. Biswal, J. R. Lee and S. Rao, *Eigenvalue bounds, spectral partitioning, and metrical deformations via flows*, J. ACM **57**(3) (2010), Article 13. doi:10.1145/1706591.1706593. Preprint version used for the intersection convention: arXiv:0808.0148v2.
- [2] J. Edmonds, *Submodular functions, matroids, and certain polyhedra*, in Combinatorial Structures and Their Applications, Gordon and Breach, 1970, 69–87.
- [3] A. Kündgen, M. J. Pelsmayer and R. Ramamurthi, *Finding minors in graphs with a given path structure*, J. Graph Theory **79** (2015), 30–47. doi:10.1002/jgt.21812. Primary preprint: arXiv:1207.6141.